

S.No	Particulars	Theory		Practical	
		Amplitude	Frequency	Amplitude	Frequency
1.	Message Signal	6	250	6	250.6 ✓
2.	Carrier Signal	15	2500	15	2512.6 ✓
3.	<u>Upper Sideband</u> (USB)				
	FM Signal				
4.	<u>Lower Sideband</u> (LSB)				

$$T_m = x_2 - x_1$$

$$4.99 - 1.00$$

$$= 3.99$$

$$T_m = x_2 - x_1$$

$$= 4.98 - 1.00$$

$$= 3.98$$

$$T_{max} = 0.00270$$

$$T_{min} = 0.000795$$

2703.2

1257.9

USB - 2703.2

Ex: 4
28/7/26

FREQUENCY MODULATION (FM) SIGNAL

GENERATION AND BANDWIDTH ANALYSIS.

AIM:

To generate and analyse a frequency modulated (FM) signal using Scilab by observing the message signal, carrier signal, and the FM waveform.

APPARATUS REQUIRED:

- * Computer / laptop
- * SCILAB.

CODE:

```
clc;  
clear;  
close;  
am = 6;  
fm = 250;  
ac = 15;  
fc = 2500;  
B = 5;  
fs = 25000;  
t = 0: 1/fs : 2/fm;  
em = am * sin(2 * % pi * fm * t);  
ec = ac * sin(2 * % pi * fc * t);
```

Calculation:

Message signal:

$$\begin{aligned}T_m &= x_2 - x_1 \\&= (4.99 \times 10^{-3}) - (1.00 \times 10^{-3}) \\&= 3.99 \times 10^{-3}\end{aligned}$$

Frequency:

$$\begin{aligned}f_m &= \frac{1}{T_m} \\&= \frac{1}{3.99 \times 10^{-3}} \\&= 250.6 \text{ Hz}\end{aligned}$$

Carrier signal:

$$\begin{aligned}T_m &= x_2 - x_1 \\&= (8.98 \times 10^{-4}) - (5.00 \times 10^{-4}) \\&= 3.98 \times 10^{-4}\end{aligned}$$

Frequency:

$$\begin{aligned}f_m &= \frac{1}{T_m} \\&= \frac{1}{3.98 \times 10^{-4}} \\&= 2512.6 \text{ Hz}\end{aligned}$$

$$e_{fm} = a * \cos((2 * \pi * f_c * t) + B * \sin(2 * \pi * f_m * t));$$

figure (1);

subplot (3, 1, 1);

plot (t, em);

x grid ();

title ("Message Signal");

x label ("time (s)");

y label ("Amplitude");

subplot ~~(t, ec);~~ (3, 1, 2);

plot (t, ec);

x grid ();

title ("carrier Signal");

x label ("time (s)");

y label ("Amplitude");

subplot (3, 1, 3);

plot (t, efm);

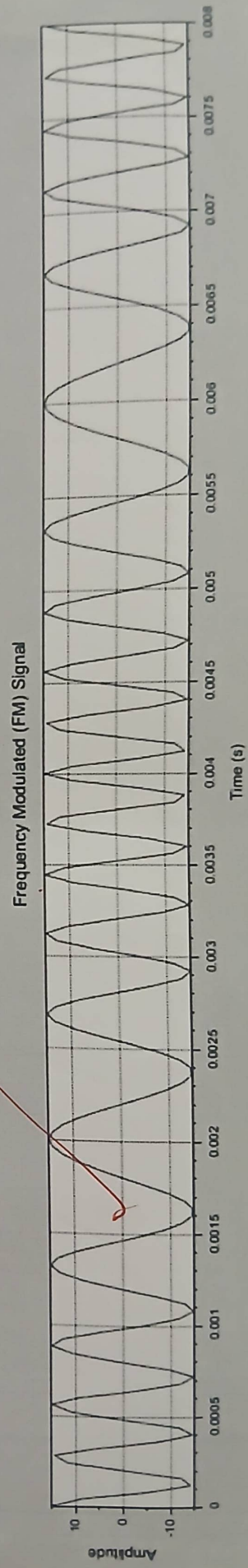
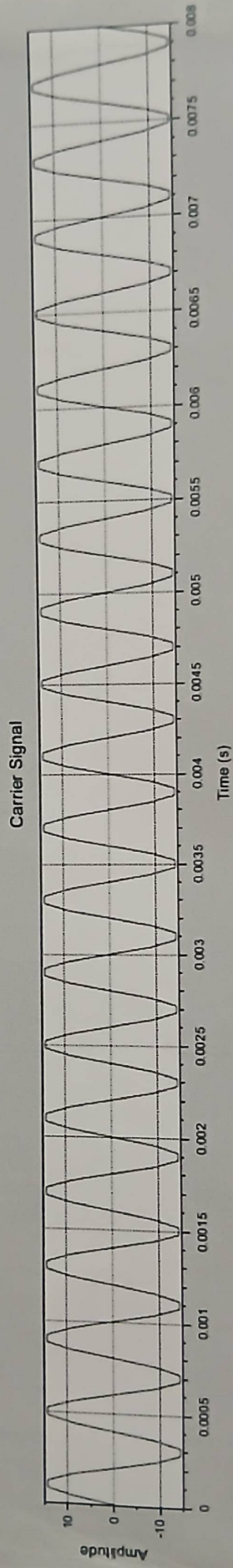
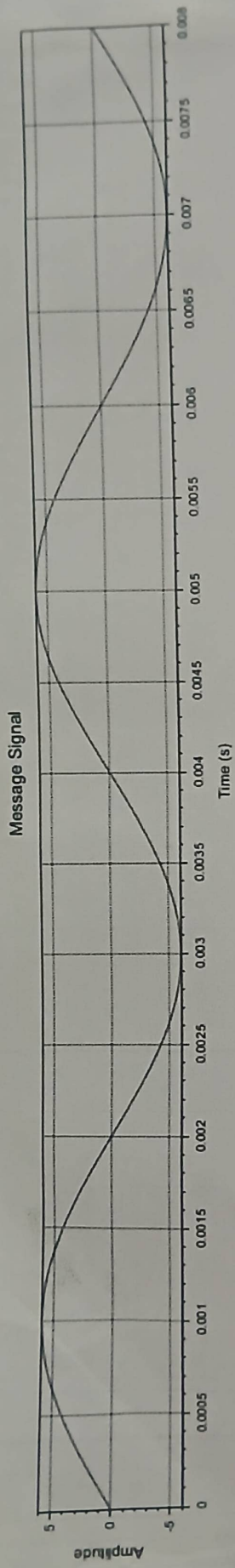
x grid ();

title ("Frequency Modulated (FM) signal");

x label ("Time (s)");

y label ("Amplitude");

FM



FM Signal:

Case 1:

$$\begin{aligned}\Delta f &= f_c - f_{\min} \\ &= 2512.6 - 1257.9 \\ &= 1254.7 \text{ Hz}\end{aligned}$$

$$\begin{aligned}\beta &= \frac{\Delta f}{f_m} \\ &= \frac{1254.7}{250} = 5.018\end{aligned}$$

Case 2:

$$\begin{aligned}\Delta f &= f_{\max} - f_c \\ &= 3703.2 - 2512.6 \\ &= 1190.6\end{aligned}$$

$$\beta = \frac{\Delta f}{f_m} = \frac{1190.6}{250} = 4.76$$

Sci lab program Implementation (20)	wave form simulation (20)	Viva (20)	Record observation (20)	Total mark
20	20	17	20	97

RESULT:

Thus the frequency Modulation (FM) signal was successfully generated using Scilab. The message signal, carrier signal and FM signal were observed.